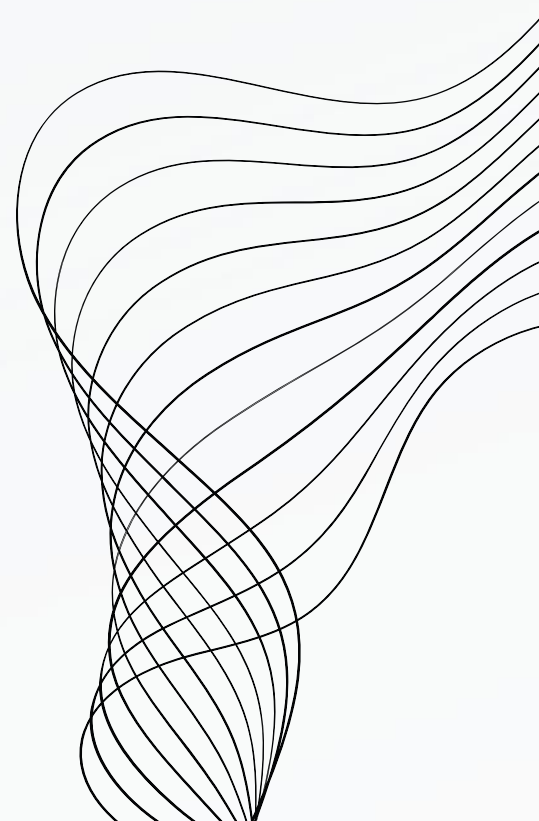


DATAVIS

LECTURE №6. INTERACTIVITY AND INTERACTION





CONTENT



01

METHODS OF INTERACTION

02

CHANGES IN TIME

03

NAVIGATION (POSITIONING)

04

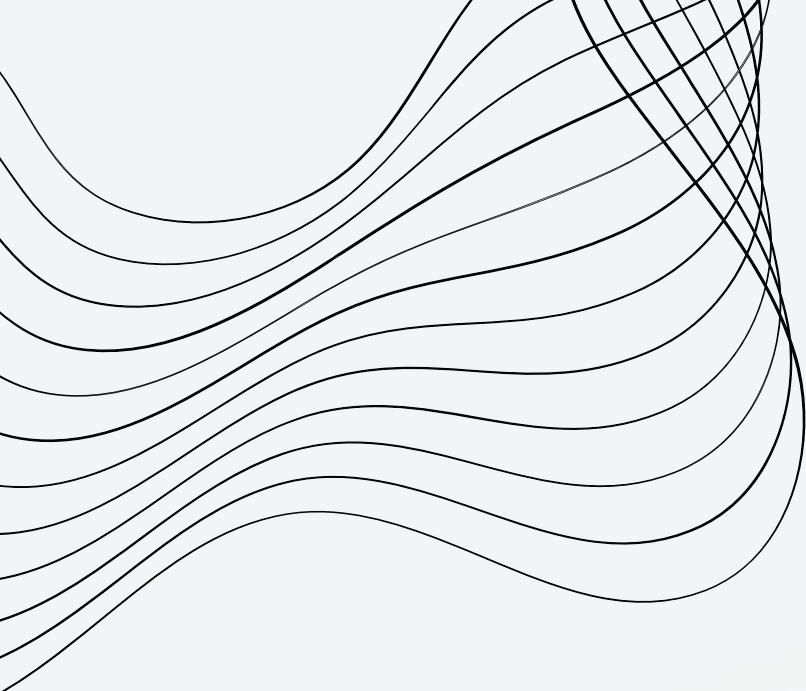
SEMANTIC ZOOM

05

FOCUS ON THE OBJECT

06

DYNAMIC QUEUE



WHY ARE INTERACTIVE VISUALIZATIONS NEEDED?

Data volume

With very large amounts of data, interactive visualization allows you to highlight key points and simplify the perception of information.

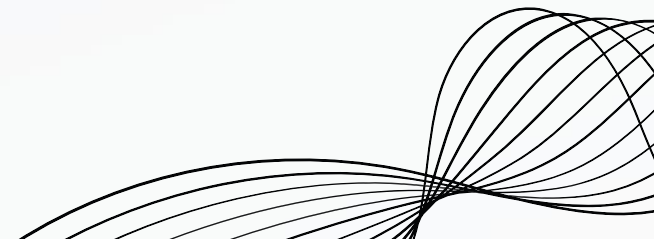
Understanding

The use of interactive visualizations stimulates not only perception but also understanding, since the viewer is directly involved in the process.





TYPES AND METHODS OF INTERACTIONS

- Single view
 - Multiple viewpoints
 - Change over time
 - Selection (Details on demand)
 - Navigation
 - Linking
 - Semantic zooming
 - Adapting representations
 - Filtering and Querying
 - Focus + Context
- 

QUOTE OF THE DAY

Jacques Bertin

"Graphics are never an end in themselves; they are a moment in the decision-making process."



French cartographer and theorist, author of a fundamental work on visual decoding: "Sémiologie Graphique"

CHANGE OVER TIME

Dynamic switching between views at different time intervals.

1. Allows you to track the dynamics;
2. Convenience of comparison;



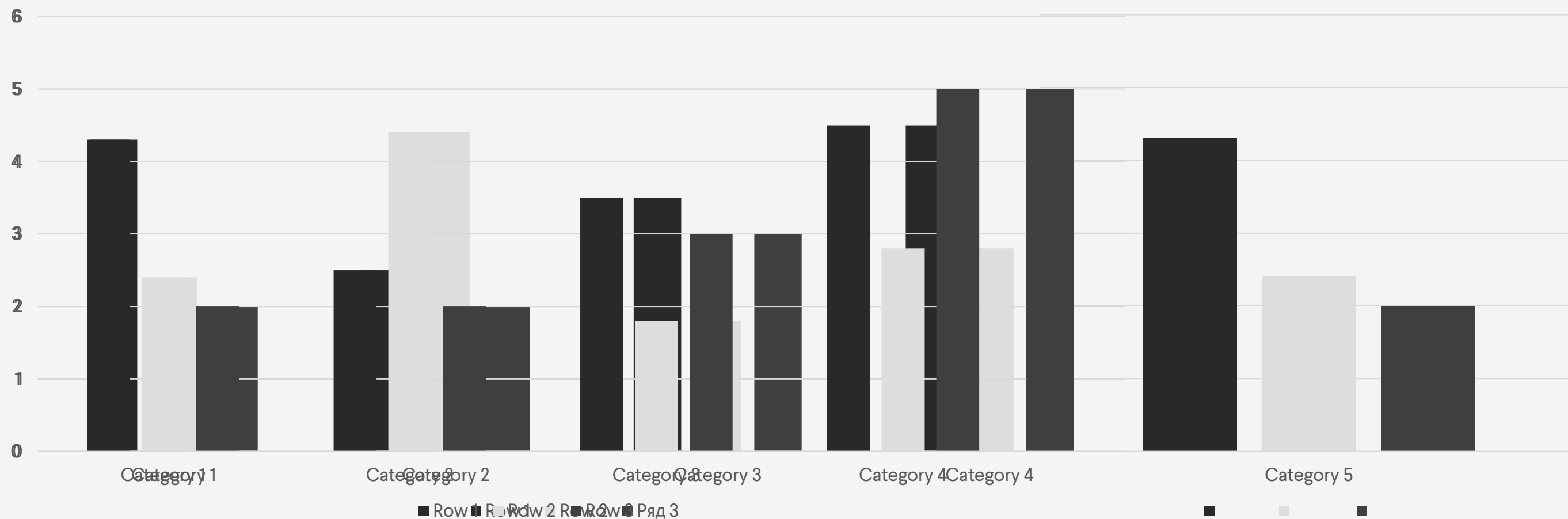
NAVIGATION (POSITIONING)



Shift and emphasis on specific objects.

1. Focuses attention;
2. Removes excess;

SEMANTIC ZOOMING



Resizing when adding new elements

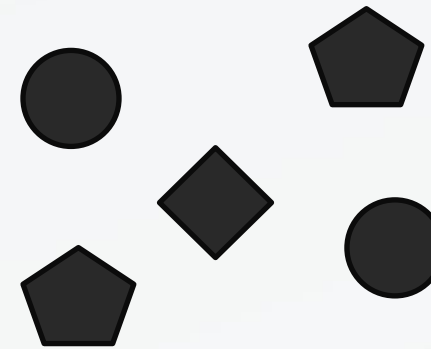
SELECTION, FILTERING AND LINKING

This is Selection



More information about object

This is filtering



And this is linking



Degree of Interest

$$DOI(x) = API(x) - D(x, y)$$

- *API* – a priori interest;
- *D* – a distance function to the current focus;

Degree-of-Interest Trees: A Component of an Attention-Reactive User Interface

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ABSTRACT

This paper proposes Degree-of-Interest trees. These trees use degree-of-interest calculations and focus+context visualization methods, together with bounding constraints, to fit within pre-established bounds. The method is an instance of an emerging “attention-reactive” user interface whose components are designed to snap together in bounded spaces.

Categories and Subject Descriptors

H.5.2 (Information Interfaces and Presentation): Graphical user interfaces. I.3.6. (Methodology and Techniques): Interaction techniques.

General Terms

Human Factors

Keywords

Degree-of-Interest Trees, DOI Trees, focus+context, information visualization, attention-reactive user interfaces, fisheye displays, hierarchical display, tree

1. INTRODUCTION

Current technology makes it feasible to bring increasingly large amounts of information to bear in computer applications. This paper explores one instance of a general strategy for constructing interfaces for high-information applications. The strategy is the Attention-reactive User Interface (AUI). Such an interface consists of two parts. One part is a method for continuous prediction of the user’s instantaneous Degree-of-Interest (DOI) over items in the field of information. The other part is a dynamic visual display of the information that uses the DOI calculation to reduce the time cost of information access or to manage attention. DOI calculations could be used to allocate display resources, decide which elements to display, change representation, highlight, or take initiative in a mixed-initiative dialogue (see Figure 1).

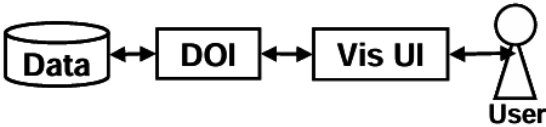


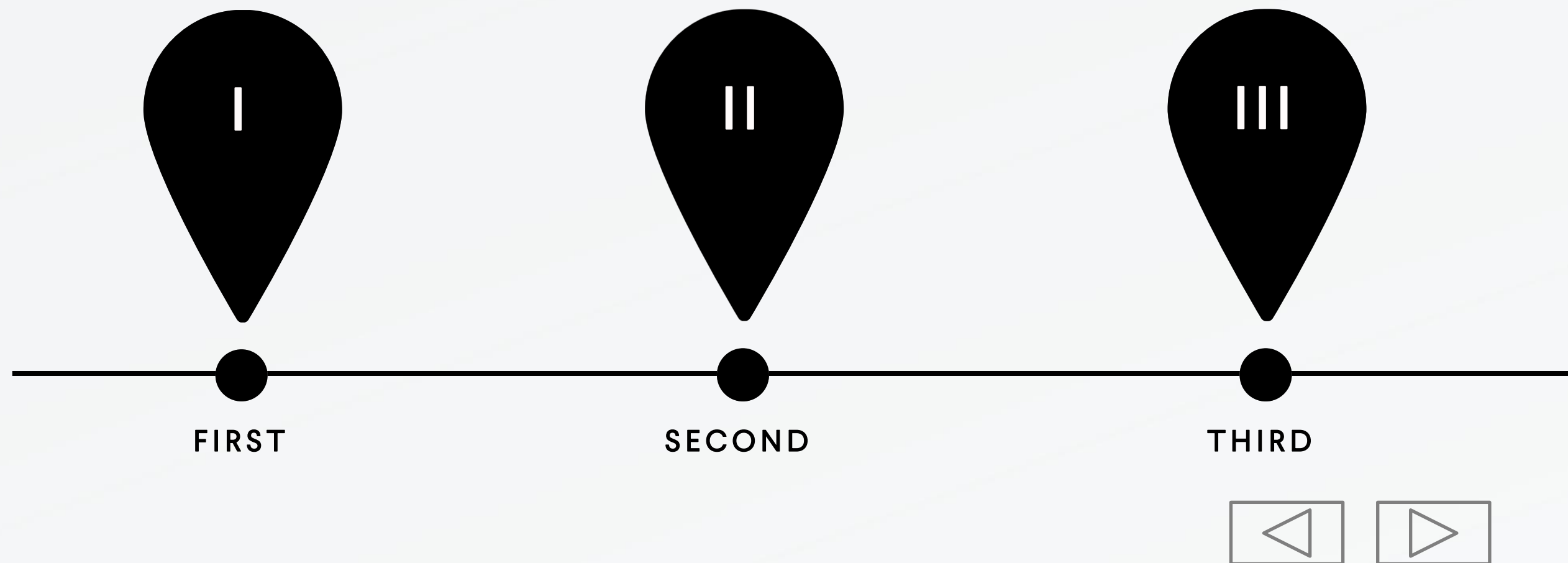
Figure 1. Attention-Reactive User Interface.

The instance of this paradigm we explore here is a method for dynamically interacting with hierarchical information in trees. Hierarchical displays are important not only because many interesting collections of information, such as organization charts or taxonomies, are hierarchical in form, but also because important collections of information, such as Websites, are *approximately* hierarchical. Whereas practical methods exist for displaying trees up to several thousand nodes, no good methods exist for displaying general graphs of this size. Hence, tree-based displays are more important than just as displays of hierarchical data.

Good visualizations of hierarchical information would (1) allow adequate space in nodes to display information, (2) allow users to understand the relationship of a node to its surrounding context, (3) allow users to find elements in the hierarchy quickly, and (4) fit into a bounded region. This last requirement is desirable in order to insure information fits on a display or that it can compose together with other display elements in an application without the need for scrolling.

For purpose of illustration, we shall use the display of organization charts as our example of hierarchical data, keeping in mind that the same techniques work for many different types of trees. Organization charts are trees (at least the ones we shall discuss first are) and have nodes that display several properties per node.

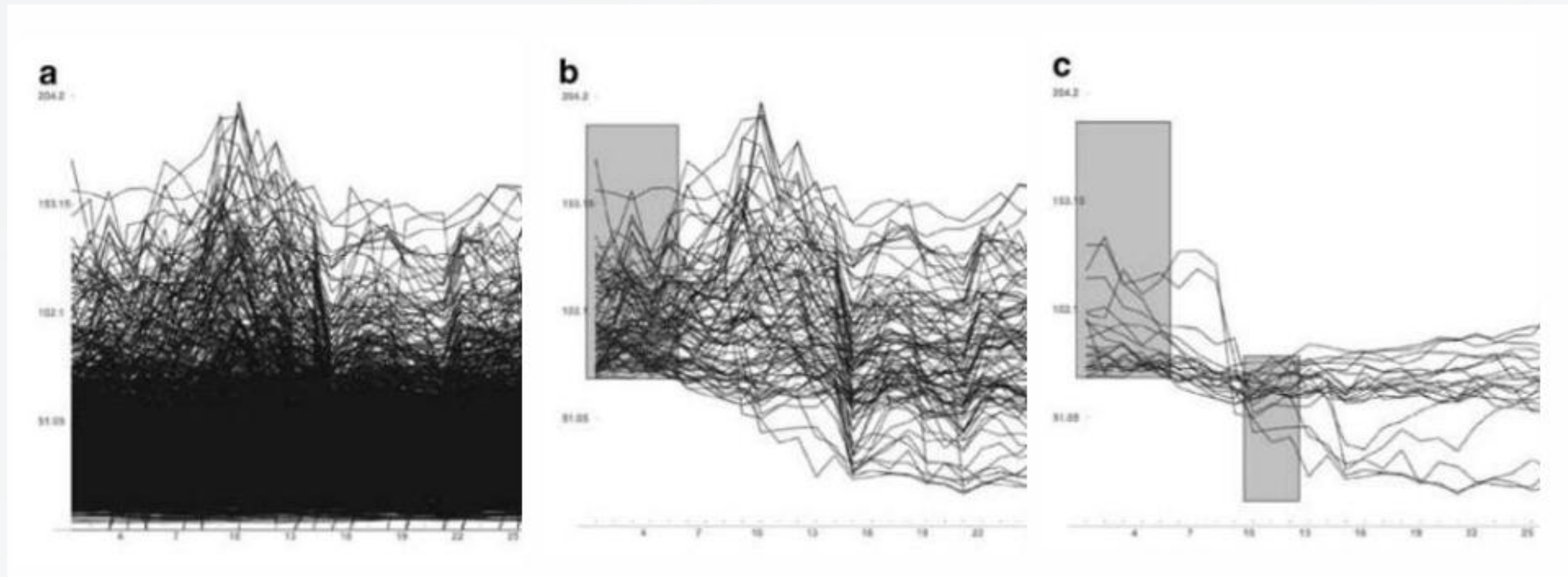
FOCUS ON THE OBJECT



Change active object.

1. Increases engagement;
2. Ability to specify viewing order or transfer chronology;

DYNAMIC VISUAL QUERIES

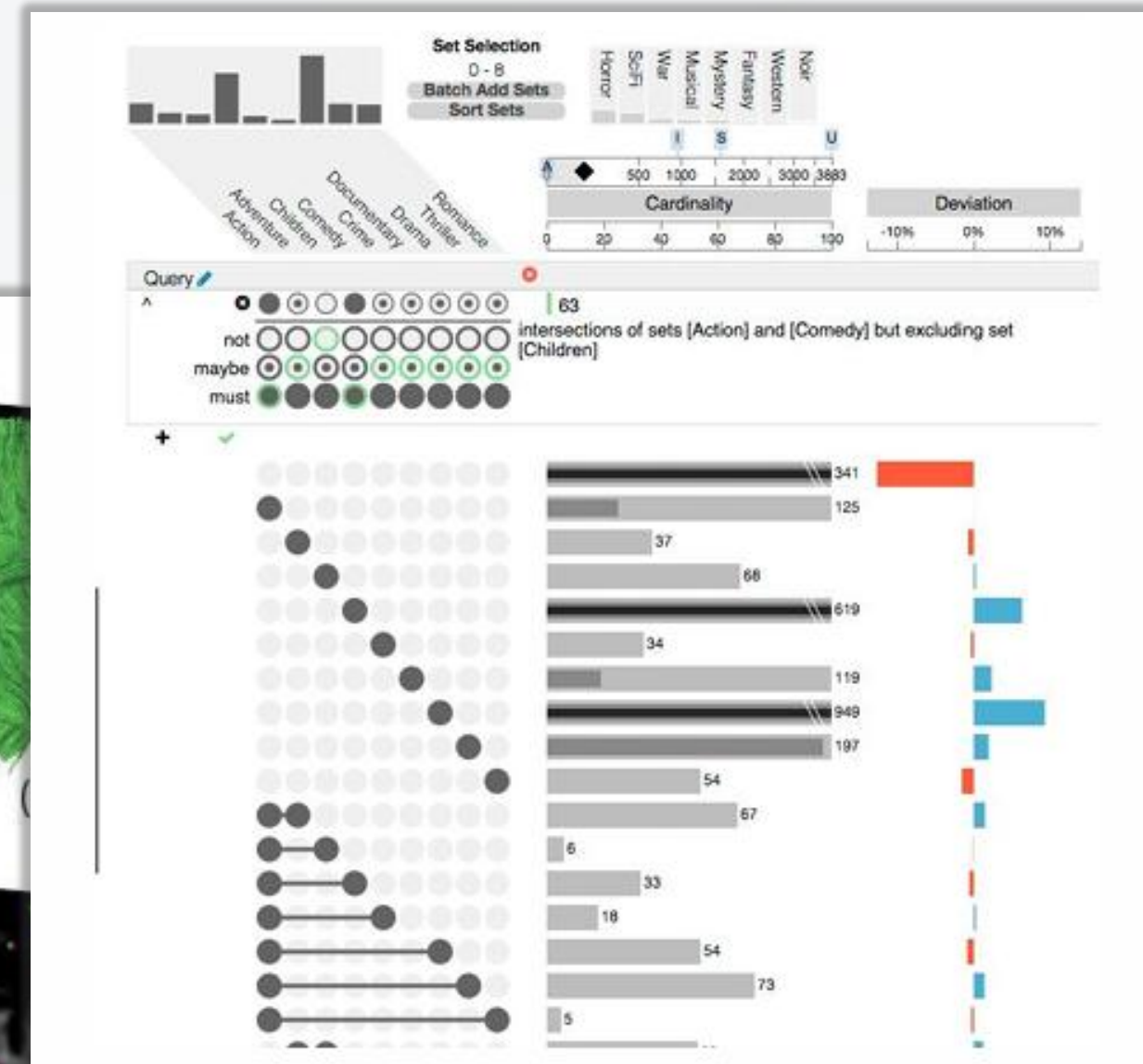
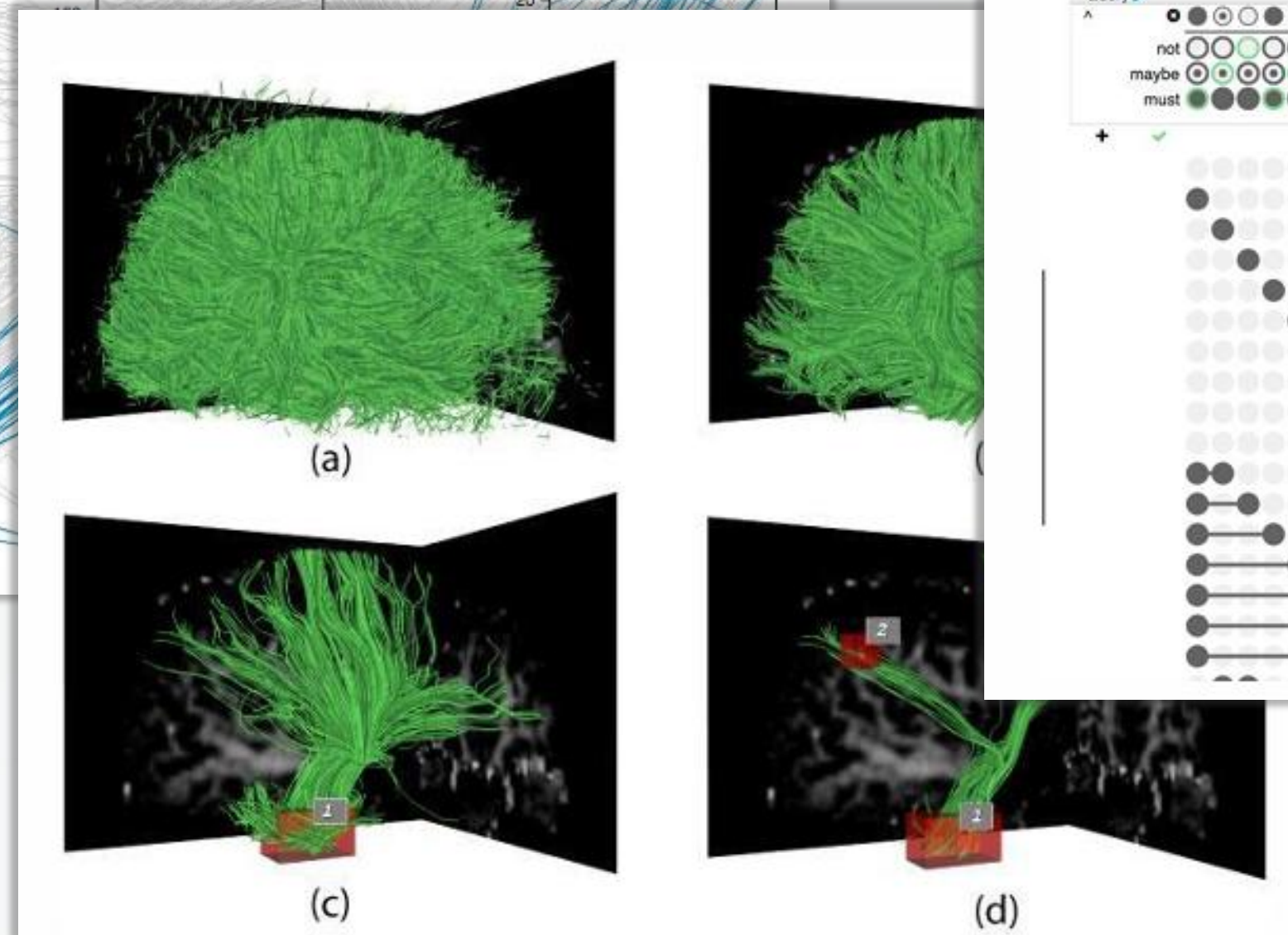
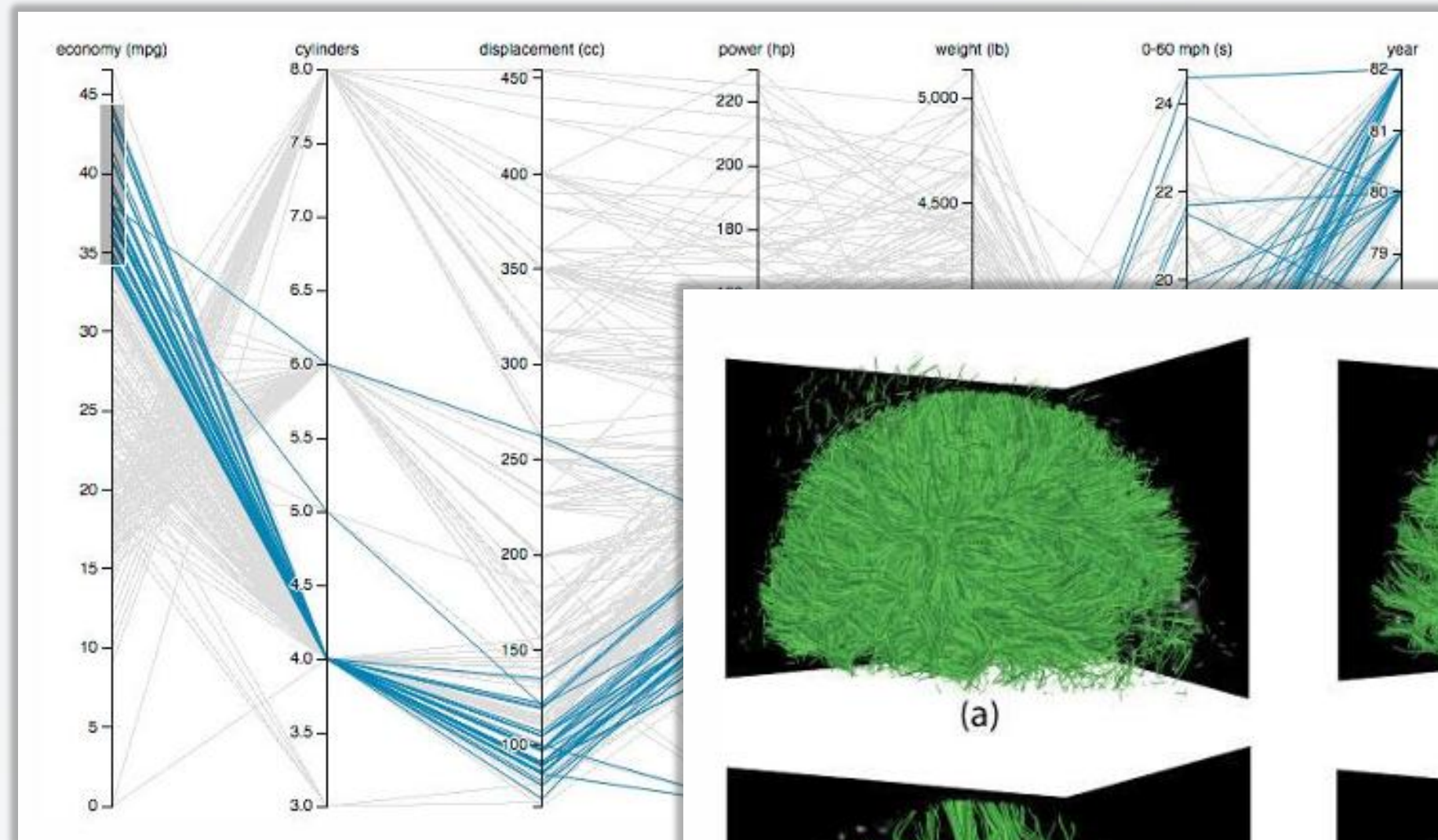


Time Searcher (Hocheiser, 2003)

Consistent change of the object of attention and addition of new ones.

1. Creates the illusion of many objects;
2. Simplifies perception due to consistency;

DYNAMIC VISUAL QUERIES





Overview first, zoom and
filter, then details on
demand.

“The Mantra of Visual Information Search.”
Shneiderman, 1996

LIST OF REFERENCES

